Notes on the continuum sigma limit

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1 Introduction

2 Green's function transformation

To start our analysis, consider a linear differential operator L acting on $\mathbb{R}^n$, and its Green's function $G(x, y)$, defined by

$$LG(x, y) = \delta(x - y). \tag{1}$$

Introduce a variable s, and define a heat kernel $K(s, x, y)$ by the equation

$$\frac{\partial}{\partial s} K(s, x, y) = LK(s, x, y), \tag{2}$$

subject to the initial conditions $K(0, x, y) = \delta(x - y)$. If the operator L is dissipative in the sense that $\lim_{s\to\infty} K(s, x, y) = 0$, we can readily prove the relation

$$G(x, y) = -\int_0^\infty K(s, x, y) ds. \tag{3}$$

To prove this relation, act with L on both sides, to find

$$LG(x, y) = -\int_0^\infty \frac{\partial}{\partial s} K(s, x, y) ds = -K(s, x, y)\Big|_0^\infty = \delta(x - y), \tag{4}$$

which proves the identity assuming that L can be interchanged with the integration. With these definitions, one can define a generalized Weierstrass transform W_s through the convolution

$$W_s f(x) = \int K(s, x, y) f(y) d^n y, \tag{5}$$

which also satisfies the heat equation

$$\frac{\partial}{\partial s} W_s f(x) = LW_s f(x), \tag{6}$$

subject to the initial conditions $W_0 f(x) = f(x)$. This leads to the relation

$$f(x) = -L \int_0^\infty W_s f(x) ds, \tag{7}$$

which is a way of writing the solution to the potentential equation $Lu(x) = f(x)$.

As an interesting special case, consider the operator L to be the Laplacian Δ acting on $\mathbb{R}^3$, for which the heat kernel is given by

$$K(s, x, y) = (4\pi s)^{-3/2} e^{-|x-y|^2/4s}. \tag{8}$$

In this case, equation (3) becomes

$$-\frac{1}{4\pi|x-y|} = -\int_0^\infty (4\pi s)^{-3/2} e^{-|x-y|^2/4s} ds, \tag{9}$$

which can be readily checked with the change of variables $s^2 = |x-y|^2/4s$. The transform W_s is then the generalized Weierstrass transform in three dimensions. If the function $f(x)$ is a charge density, denoted $\rho(x)$, then we find

$$V(x) = \frac{1}{\varepsilon_0} \int_0^\infty W_s \rho(x) ds, \tag{10}$$

where $V(x)$ is a solution of Poisson's equation

$$\Delta V(x) = -\frac{\rho(x)}{\varepsilon_0}, \tag{11}$$

and ε_0 is the vacuum permittivity.

We will be interested in the application of this identity to quantum mechanical systems. In that context, it is relevant to write the generalized Weierstrass transform in the form

$$W_s = e^{sL}, \tag{12}$$

which follows formally from equation (6). It will be desirable to define an operator $\hat{L}$ which performs a Weierstrass transform through conjugation

$$W_s \hat{A} = e^{s\hat{L}} \hat{A} e^{-s\hat{L}}, \tag{13}$$

which would follow formally from the identity [1]

$$e^X Y e^{-X} = Y + [X, Y] + \frac{1}{2!}[X, [X, Y]] + \frac{1}{3!}[X, [X, [X, Y]]] + \dots. \tag{14}$$

Without going into details, there are some difficulties with this procedure if L is chosen to be the Laplacian and the operator $\hat{A}$ is chosen to be the particle density, which makes this identity

less easily applied in that context. For that reason, we will also develop analogous formulas for first order Dirac operators, which do have the properties outlined above.

With this motivation, we introduce a Euclidean Clifford algebra in $\mathbb{R}^n$ with a basis $(e_1, \ldots, e_n)$ satisfying

$$\{e_i, e_j\} = -2\delta_{ij}, \tag{15}$$

and define the Euclidean Dirac operator

$$\not{\partial} = \sum_{i=1}^{n} e_i \partial_i, \tag{16}$$

which satisfies

$$\not{\partial}\not{\partial} = -\Delta. \tag{17}$$

With this notation, Poisson's equation $\Delta u(x) = -f(x)$ can be recast as the system of equations

$$\not{\partial} v(x) = f(x), \quad \not{\partial} u(x) = v(x). \tag{18}$$

As before, we can define a Green's function and kernel for this operator, with a subscript to avoid confusion

$$\not{\partial} G_{\frac{1}{2}}(x, y) = \delta(x - y), \tag{19}$$

$$\frac{\partial}{\partial s} K_{\frac{1}{2}}(s, x, y) = \not{\partial} K_{\frac{1}{2}}(s, x, y), \tag{20}$$

with the initial condition $K_{\frac{1}{2}}(0, x, y) = \delta(x - y)$ for the kernel. As before, if we assume the operator is dissipative, we have the identity

$$G_{\frac{1}{2}}(x, y) = -\int_0^\infty K_{\frac{1}{2}}(s, x, y) ds. \tag{21}$$

In this case, the assumption of the operator being dissipative is more subtle, and will be discussed in more depth with some examples.

Using these equations, we can formally write down a "first order" solution to Poisson's equation. First, solve for $u(x)$ and $v(x)$ using the method outlined previously

$$v(x) = -\int_0^\infty ds \int dy K_{\frac{1}{2}}(s, x, y) f(y), \tag{22}$$

$$u(x) = -\int_0^\infty ds \int dy K_{\frac{1}{2}}(s, x, y) v(y), \tag{23}$$

and then combine these equations to find

$$u(x) = \int_0^\infty ds\, dt \int dy\, dz\, K_{\frac{1}{2}}(s, x, y) K_{\frac{1}{2}}(t, y, z) f(z). \tag{24}$$

Using the propagator identity

$$K_{\frac{1}{2}}(s+t,x,z) = \int dy K_{\frac{1}{2}}(s,x,y)K_{\frac{1}{2}}(t,y,z), \tag{25}$$

in addition to the change of variables $s = s + t$ and $\sigma = s - t$, we find

$$u(x) = \frac{1}{2}\int_0^\infty ds \int_{-s}^{s} d\sigma \int dy\, K_{\frac{1}{2}}(s,x,y)f(y) = \int_0^\infty ds \int dy\, sK_{\frac{1}{2}}(s,x,y)f(y). \tag{26}$$

We emphasize that this formula is only correct under the assumption that the operator is dissipative, which is not true in general for first order operators, and will be discussed in more depth with a few examples.

3 Dirac operators in various dimensions

To gain some more intuition for when equation (26) is valid, we first consider a one dimensional system, which is the simplest example. In this case, the Dirac operator is $\not{\partial} = e_1\partial_x$, and the equation for the Dirac kernel is

$$\Big(\frac{\partial}{\partial s} - e_1\partial_x\Big)K_{\frac{1}{2}}(s,x,y) = 0. \tag{27}$$

With the form of this equation, it is clear that this can be mapped to the Cauchy-Riemann equations. With this in mind, we define the complex number $z = s - ix$, identifying i with e_1, so that equation (27) does give the Cauchy-Riemann equations. To find the solution for the kernel, we have to impose the initial conditions, which are $K_{\frac{1}{2}}(0,x,y) = \delta(x-y)$. Using the Fourier transform of the Dirac delta function, we have, formally

$$K_{\frac{1}{2}}(0,x,y) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i(x-y)\xi}d\xi. \tag{28}$$

For any individual Fourier component of this integral, the solution to the Cauchy-Riemann equations with the given initial condition is $e^{-(z+iy)\xi}$, which formally leads to

$$K_{\frac{1}{2}}(s,x,y) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-(s-ix+iy)\xi}d\xi. \tag{29}$$

It is now clear why the dissipative assumption is non-trivial for a first order operator, because these solutions are only dissipative for positive s if $\xi > 0$.

It is clear from the above discussion that equation (26) is not valid for all functions $f(y)$. However, there is a way of modifying the equation to become valid. To see this, first define positive and negative Dirac kernels as follows

$$K_+(s,x,y) = \frac{1}{2\pi}\int_0^\infty e^{-(s-ix+iy)\xi}d\xi, \quad K_-(s,x,y) = \frac{1}{2\pi}\int_{-\infty}^0 e^{-(s-ix+iy)\xi}d\xi, \tag{30}$$

so that the full kernel can formally be written $K_{\frac{1}{2}}(s,x,y) = K_+(s,x,y) + K_-(s,x,y)$. A short calculation shows that

$$K_+(s,x,y) = \frac{1}{2\pi(s-ix+iy)}, \quad s>0, \tag{31}$$

$$K_-(s,x,y) = \frac{-1}{2\pi(s-ix+iy)}, \quad s<0, \tag{32}$$

where the integrals do not converge outside the domain of defition. With this in mind, the correct analogue to equation (21) is

$$G_{\frac{1}{2}}(x,y) = -\int_0^\infty K_+(s,x,y)ds + \int_{-\infty}^0 K_-(s,x,y)ds, \tag{33}$$

which can again be proved by operating with $\not{\partial}$ as in equation (4). This can be rewritten

$$G_{\frac{1}{2}}(x,y) = -\int_{-\infty}^\infty K_{\frac{1}{2}}(s,x,y)ds, \tag{34}$$

where $2\pi K_{\frac{1}{2}}(s,x,y) = (s-ix+iy)^{-1}$ is defined for all s. We note that this is the fundamental solution for the Cauchy-Riemann operator in the variable $z = s-ix$, which will be important for the generalization to higher dimensions[1]. Using contour integration, one can readily compute this integral to find

$$-\int_{-\infty}^\infty K_{\frac{1}{2}}(s,x,y)ds = -iH(x-y) + \frac{i}{2}, \tag{35}$$

where $H(x-y)$ is the Heaviside step function. This is indeed the Green's function of the operator $\not{\partial} = i\partial_x$, which is a reasurring consistency check for the method outlined.

Proceeding as before, the solution to the equation $\not{\partial}v(x) = f(x)$ can be written

$$v(x) = -\int_{-\infty}^\infty ds \int dy K_{\frac{1}{2}}(s,x,y)f(y). \tag{36}$$

[1]There is a factor of two relative to the usual definition of the Cauchy kernel, which comes from the definitions used in the integration measure.

From equation (26), one might guess that a solution to Poisson's equation $\not{\partial} u(x) = v(x)$ can be written

$$u(x) = \int_{-\infty}^{\infty} ds \int dy\, s K_{\frac{1}{2}}(s, x, y) f(y). \tag{37}$$

This can be proved formally by operating with $\not{\partial}$ and integrating by parts

$$\begin{aligned} \not{\partial} u(x) = \int_{-\infty}^{\infty} ds \int dy\, s \frac{\partial}{\partial s} K_{\frac{1}{2}}(s, x, y) f(y) = \\ - \int_{-\infty}^{\infty} ds \int dy\, K_{\frac{1}{2}}(s, x, y) f(y) = v(x), \end{aligned} \tag{38}$$

where the boundary terms cancel. However, this integral may diverge, and the validity of this proof depends on the function $f(y)$. The integral over s does converge in higher dimensions, however, which will be shown explicitly for $n = 3$.

The one dimensional case is a useful illustration of the general situation in higher dimensions, which is analogous. For general n, the (massless) Dirac kernel is defined by equation (20), and we will consider a general represenataion of the Clifford algebra on which the Dirac operator acts. The relevant exponential solutions of this equation take the form

$$\psi^{\pm}(x, y; p) = u^{\pm}(p) e^{\mp Es + ip\cdot(x-y)}, \tag{39}$$

where we have defined a positive "energy" satisfying $E^2 = p^2$, using $\pm$ to denote the s direction in which the solution decays. Here $u^{\pm}(p)$ is a spinor satisfying

$$(E \pm i\not{p}) u^{\pm}(p) = 0. \tag{40}$$

This equation has multiple solutions, so we will introduce an index α to distinguish the different solutions. Define the projectors

$$\pi_{\pm}(p) = \frac{1}{2E}\big(E \mp i\not{p}\big). \tag{41}$$

These are projectors because they satisfy $\pi_{\pm}(p)^2 = \pi_{\pm}(p)$, and they are complementary because $\pi_{+}(p) = 1 - \pi_{-}(p)$. The solutions $u^{\pm}(p)$ are in the positive and negative subspaces of these projectors. Defining $u^{\pm}_{\alpha}(p)$ as bases for these subspaces, we choose the normalization condition $\bar{u}^{\pm}_{\alpha}(p) u^{\pm}_{\alpha}(p) = 1$, which leads to the completeness relations[2]

$$\sum_{\alpha} u^{\pm}_{\alpha}(p) \bar{u}^{\pm}_{\alpha}(p) = \pi_{\pm}(p). \tag{42}$$

[2]Because we are considering Euclidean space, the Dirac adjoint $\bar{u}$ is the same as the Hermitian adjoint $u^{\dagger}$.

As before, using the Fourier transform of the delta function $(2\pi)^n\delta(x-y) = \int e^{i(x-y)\xi}d^n\xi$, we formally find

$$K_{\frac{1}{2}}(s,x,y) = \frac{1}{(2\pi)^n}\sum_{\pm}\sum_{\alpha}\int d^n p\, u_\alpha^{\pm}(p)\bar{u}_\alpha^{\pm}(p)e^{\mp Es+ip\cdot(x-y)}, \tag{43}$$

where here we have used the total completeness relation $\sum_{\pm}\sum_{\alpha} u_\alpha^{\pm}(p)\bar{u}_\alpha^{\pm}(p) = 1$. Define the positive and negative kernels by the equations

$$K_{\pm}(s,x,y) = \frac{1}{(2\pi)^n}\int d^n p\, \pi_{\pm}(p)e^{\mp Es+ip\cdot(x-y)}. \tag{44}$$

The domains of convergence for these integrals are again positive and negative s.

With this formalism, equation (33) can be written as in the one dimensional case

$$G_{\frac{1}{2}}(x,y) = -\int_0^{\infty} K_+(s,x,y)ds + \int_{-\infty}^{0} K_-(s,x,y)ds, \tag{45}$$

with the same proof. This can again be written

$$G_{\frac{1}{2}}(x,y) = -\int_{-\infty}^{\infty} K_{\frac{1}{2}}(s,x,y)ds, \tag{46}$$

where $K_{\frac{1}{2}}(s,x,y)$ is the fundamental solution of the operator $\frac{\partial}{\partial s} - \not{\partial}$. To see this, consider a function $f(x)$, and note that the convolution $v(x) = \int G_{\frac{1}{2}}(x,y)f(y)d^n y$ is a solution to the equation $\not{\partial}v(x) = f(x)$. Operating with $\not{\partial}$ on $v(x)$ after using the above equation, we find

$$\not{\partial}v(x) = -\int_{-\infty}^{\infty} ds\int d^n y\Big(\frac{\partial}{\partial s}K_{\frac{1}{2}}(s,x,y)f(y) - \delta^{n+1}(s,x-y)f(y)\Big) = f(x), \tag{47}$$

which is the desired result. Here, $\delta^{n+1}(s,x-y)$ denotes the delta function where the s dimension is written separately, and the boundary terms vanish for a well-behaved function $f(y)$. With this result, analogues of equations (36) and (37) also follow with the same arguments as before.

The fundamental solution of the operator $\frac{\partial}{\partial s} - \not{\partial}$ can be found by considering an $n+1$-dimensional Clifford algebra, where the first dimension is s. The Dirac operator is

$$\not{\partial}_{n+1} = e_0\frac{\partial}{\partial s} + \sum_{i=1}^{n} e_i\partial_i, \tag{48}$$

with the fundamental solution [2]

$$G_{\frac{1}{2}}^{(n+1)}(s,x,y) = \frac{1}{\omega_{n+1}}\frac{e_0 s + \sum_i e_i(x_i-y_i)}{|e_0 s + \sum_i e_i(x_i-y_i)|^{n+1}}, \tag{49}$$

where ω_{n+1} is the surface area of the unit sphere in $\mathbb{R}^{n+1}$, and we have written the expression explicitly for the purpose of calculation. Right multiplying equation (48) by $-e_0$ gives

$$-\not{\partial}_{n+1}e_0 = \frac{\partial}{\partial s} - \sum_{i=1}^{n} e_i e_0 \partial_i. \tag{50}$$

The products $e_i e_0$ form a representation of the Clifford algebra in dimension n, so we can redefine them as e_i, turning this into the desired operator $\frac{\partial}{\partial s} - \not{\partial}_n$. Left multiplying $G^{(n+1)}_{\frac{1}{2}}(s,x,y)$ by e_0 and rearranging, we find

$$K^{(n)}_{\frac{1}{2}}(s,x,y) = -\frac{1}{\omega_{n+1}} \frac{s + \not{x} - \not{y}}{(s^2 + |\not{x} - \not{y}|^2)^{(n+1)/2}}, \tag{51}$$

which is the desired expression.

For the application to quantum mechanics, we will focus on dimension three, so it will be useful to work out some of the details in this dimension. In this case, the kernel is given by

$$K_{\frac{1}{2}}(s,x,y) = -\frac{1}{2\pi^2} \frac{s + \not{x} - \not{y}}{(s^2 + |\not{x} - \not{y}|^2)^2}, \tag{52}$$

and the Green's function is given by

$$G_{\frac{1}{2}}(x,y) = \frac{1}{4\pi} \frac{\not{x} - \not{y}}{|\not{x} - \not{y}|^3}, \tag{53}$$

so that equation (46) becomes

$$\frac{1}{4\pi} \frac{\not{x} - \not{y}}{|\not{x} - \not{y}|^3} = \int_{-\infty}^{\infty} \frac{1}{2\pi^2} \frac{s + \not{x} - \not{y}}{(s^2 + |\not{x} - \not{y}|^2)^2} ds, \tag{54}$$

which can be verified directly[3]. The solution of Poisson's equation can be written using equation (37)

$$u(x) = \int_{-\infty}^{\infty} ds \int d^3y \, s K_{\frac{1}{2}}(s,x,y) f(y), \tag{55}$$

which leads to the following formula for the Green's function of the Laplacian

$$G(x) = \int_{-\infty}^{\infty} s K_{\frac{1}{2}}(s,x,y) ds, \tag{56}$$

This equation can also be verified directly in this case, and will be a relevant result for the application to quantum mechanics.

[3]The author used Wolfram Alpha for this purpose.

4 Application to quantum mechanics

These identities are interesting in their own right, and it is likely that there are multiple ways to apply them in a variety of contexts. Here we will apply this identity to the Coulomb potential in the nonrelativistic molecular Hamiltonian, and use a diagonalization procedure similar to the flow equation approach introduced by Wegner to estimate the correlation energy and low lying excitations in the system. To start, write the molecular Hamiltonian in the form $\hat{H} = \hat{H}_0 + \hat{V}_{ee}$, where

$$\hat{H}_0 = \int d^3x \, a^\dagger_\sigma(x) \Big(\frac{\hat{p}^2}{2m} + V(x) \Big) a_\sigma(x). \tag{57}$$

Here σ is a spin index, where repeated indices are summed, and $V(x)$ includes the ionic potential, which will be treated as approximately fixed. $\hat{V}_{ee}$ is the second quantized Coulomb potential

$$\hat{V}_{ee} = 2\pi e^2 \int d^3x \int d^3y \frac{1}{4\pi|x-y|} a^\dagger_\sigma(x) a^\dagger_{\sigma'}(y) a_{\sigma'}(y) a_\sigma(x). \tag{58}$$

To make contact with the previous analysis, define a heat kernel operator by

$$\hat{K}(s) = \int d^3x \int d^3y K(s,x,y) a^\dagger_\sigma(x) a^\dagger_{\sigma'}(y) a_{\sigma'}(y) a_\sigma(x). \tag{59}$$

Equation (9) then readily implies

$$\hat{V}_{ee} = 2\pi e^2 \int_0^\infty ds \hat{K}(s). \tag{60}$$

We will attempt to approximately diagonalize the full Hamiltonian with a unitary flow. To do this, we first define a reference state, which will be used to define normal ordering for the system. An explicit example of this normal ordering procedure will be considered with a Slater determinant reference state. We will then systematically diagonalize each infinitesimal contribution to the integral in equation (71), making a strong physical approximation to truncate the flow.

A unitary flow is generated by an antihermitian operator $\eta(s)$ through the equation

$$\frac{d\hat{O}(s)}{ds} = [\eta(s), \hat{O}(s)], \tag{61}$$

which is true of any operator $\hat{O}(s)$. In particular, if one takes the creation and annihilation operators, this generates a canonical transformation for the many-particle Hilbert space. If the operator $\hat{O}(s)$ is chosen to be the Hamiltonian, this equation determines the structure of the

flowing Hamiltonian, which will be the central object of study in this analysis. In particular, we will write the Hamiltonian in the form

$$\hat{H}(s) = E(s) + \hat{H}_0(s) + \int_s^\infty ds' \hat{K}(s') + o(m). \tag{62}$$

Here $E(s)$ is the ground state energy of the flowing Hamiltonian, which will approach the desired ground state energy as $s \to \infty$. $\hat{H}_0(s)$ is the diagonal part of the Hamiltonian, and $o(m)$ corresponds to the terms that will be dropped. We will attempt to systematically diagonalize the interaction with the unitary flow, by diagonalizing each term $\hat{K}(s')$ as a function of s'.

In order to successively diagonalize the Hamiltonian as a function of s, we will want to choose the generator $\eta(s)$ such that

$$[\eta(s), \hat{H}_0(s)] = \frac{\partial}{\partial s}\Big(E(s) + \hat{H}_0(s)\Big) + \hat{K}(s). \tag{63}$$

Furthermore, we will demand that the the operator $\eta(s)$ leaves the creation and annihilation operators invariant to first order in the normal ordered expression, so that

$$: [\eta(s), a_\sigma(x)] :_T = 0, \quad \forall x, \sigma. \tag{64}$$

The notation :: denotes normal ordering with respect to the reference state, and the subscript T denotes the act of truncating the terms that contain more than one creation or annihilation operator in the normal ordering. Physically this means that, with respect to the reference state, there is no forst order change in the basis of operators. These conditions will uniquely determine the structure of $\eta(s)$. If this equation is satisfied, then the Hamiltonian will be diagonalized up to order m as $s \to \infty$. Additionally, we can derive the total change in $\hat{K}(s)$,

$$\frac{d\hat{K}(s)}{ds} = [\eta(s), \hat{K}(s)] + \frac{\partial \hat{K}(s)}{\partial s}, \tag{65}$$

where the partial derivative does not take into account the change in the basis of operators. This equation reflects the fact that $\hat{K}(s)$ is a function of the parameter s as well as the basis of operators parameterized by s.

The physical approximation that we will make to drastically simplify the flow equations is

$$[\eta(s), \hat{K}(s)] \approx 0. \tag{66}$$

What this approximation means physically is that the unitary transformations that diagonalize the different length scales are effectively uncoupled, and leave the interaction terms from the different length scales form invariant. If this is the case, then $\hat{K}(s)$ does not flow, and so the equations are fully truncated. Note that it is important to impose the condition (64) in order to make this physical approximation, because otherwise there would be a first order change in the operators, which would make the form invariance an unreasonable approximation.

With this physical approximation, the operator $\eta(s)$ can always be chosen to be an operator of fourth order, and the flow of the noninteracting Hamiltonian $E(s) + \hat{H}_0(s)$ can be computed. To do this, we change to the eigenbasis of the noninteracting Hamiltonian, so that

$$\hat{H}_0 = \sum_\alpha E_\alpha a_\alpha^\dagger a_\alpha, \quad \hat{H}_{int}(s) = \sum_{\alpha\beta\gamma\delta} w_{\alpha\beta\gamma\delta}(s) a_\alpha^\dagger a_\beta^\dagger a_\gamma a_\delta. \tag{67}$$

In order to diagonalize the interacting for a particular value of s, we apply a unitary transformation generated by

$$\eta(s) = \sum_{\alpha\beta\gamma\delta} \frac{w_{\alpha\beta\gamma\delta}(s)}{\epsilon_\alpha + \epsilon_\beta - \epsilon_\gamma - \epsilon_\delta} a_\alpha^\dagger a_\beta^\dagger a_\gamma a_\delta. \tag{68}$$

The form of this term is familiar from Møller-Plesset perturbation theory, and is chosen to satisfy $[\eta(s), \hat{H}_0] = \hat{H}_{int}(s)$.

Let us derive the equations for this flow with the triple excitations ignored, and for a reference state given by a Slater determinant. In order to write the normal ordering explicitly, we will introduce a canonical basis of operators $\{a_\alpha, a_\beta^\dagger\}$ such that the the reference state is annihilated by all the operators a_α, and the Hamiltonian can be written

$$\hat{H}(0) = \sum_{\alpha\beta} H^0_{\alpha\beta} a_\alpha^\dagger a_\beta + \sum_{\alpha\alpha'\beta'\beta} \int_0^\infty ds K_{\alpha\alpha'\beta'\beta}(s) a_\alpha^\dagger a_{\alpha'}^\dagger a_{\beta'} a_\beta, \tag{69}$$

where the terms $K_{\alpha\alpha'\beta'\beta}(s)$ are determined by writing the operator $\hat{K}(s)$ in this basis.

Green's function operator for the Laplacian

$$\hat{G}(x) = -\int d^3y \frac{1}{4\pi|x-y|} \hat{\rho}(y), \tag{70}$$

so that the Coulomb operator can be written

$$\hat{V}_{ee} = -2\pi e^2 \int d^3x a_\sigma^\dagger(x) \hat{G}(x) a_\sigma(x). \tag{71}$$

Define the operator

$$\hat{K}_{\frac{1}{2}}(s,x) = \int d^3y K_{\frac{1}{2}}(s,x,y) \hat{\rho}(y), \tag{72}$$

with a similar definition for $\hat{K}(s,x)$. With these definitions, we can write operatorial generalizations of equations (3) and (56)

$$\hat{G}(x) = -\int_0^\infty \hat{K}(s,x) ds, \quad \hat{G}(x) = \int_{-\infty}^\infty s \hat{K}_{\frac{1}{2}}(s,x) ds. \tag{73}$$

$$\hat{G} = \int d^3x \Big(\int_{-\infty}^{\infty} ds \hat{K}_{\frac{1}{2}}(s,x) \Big) \Big(\int_{-\infty}^{\infty} d\tilde{s} \hat{K}_{\frac{1}{2}}(\tilde{s},x) \Big). \tag{74}$$

Define

$$\hat{G}_{\frac{1}{2}}(x) = \int_{-\infty}^{\infty} ds \hat{K}_{\frac{1}{2}}(s,x). \tag{75}$$

Then we have

$$\hat{V}_{ee} = 2\pi e^2 \int d^3x \, \hat{G}^{\dagger}_{\frac{1}{2}}(x) \hat{G}_{\frac{1}{2}}(x). \tag{76}$$

We would like to vary the Coulomb potential at fixed density and kinetic energy, which means minimizing the functional

$$\mathcal{F}(\psi) = \int d^3x \, \langle \psi | \left(2\pi e^2 \hat{G}^{\dagger}_{\frac{1}{2}}(x) \hat{G}_{\frac{1}{2}}(x) + \nu(x) \hat{\rho}(x) + \lambda(x) a^{\dagger}_{\sigma}(x) \Delta a_{\sigma}(x) \right) | \psi \rangle \, , \tag{77}$$

where $\nu(x)$ and $\lambda(x)$ are Lagrange multipliers. We would like this "energy density" to take the form a square, of the form

$$\mathcal{F}(\psi) = \int d^3x \, \langle \psi | \, \hat{F}^{\dagger}(x) \hat{F}(x) \, | \psi \rangle \, . \tag{78}$$

It turns out that this is true if $\hat{F}(x)$ takes the form

$$\hat{F}(x) = \hat{G}_{\frac{1}{2}}(x) + \sqrt{\nu} a(x). \tag{79}$$

These identities are interesting in their own right, and it is likely that there are multiple ways to apply them in a variety of contexts. Here we will use these identities to estimate the correlation energy that appears in density functional theory. To proceed, let us assume that we have a wavefunction $|\psi_E\rangle$ that is approximately an eigenfunction of the operator $\hat{V}_{ee}(x)$ at fixed density $\rho(x)$. This means that it minimizes the Coulomb energy $\langle \hat{V}_{ee}(x) + E_{ee}(x) \rho(x) \rangle_{\psi_E}$, where $E_{ee}(x)$ are a set of Lagrange multipliers. Varying this energy gives the equation

$$\hat{V}_{ee}(x) \, | \psi_E \rangle \approx \int d^3x E_{ee}(x) \hat{\rho}(x) \, | \psi_E \rangle \, . \tag{80}$$

If this equation were to hold, then the eigenvalue equation for the full Hamiltonian would become

$$\hat{H}_{eff} \, | \psi_E \rangle = E \, | \psi_E \rangle \, , \tag{81}$$

where the effective Hamiltonian is given by

$$\hat{H}_{eff} = \int d^3x \, a^{\dagger}_{\sigma}(x) \Big(\frac{\hat{p}^2}{2m} + V(x) + E_{ee}(x) \Big) a_{\sigma}(x). \tag{82}$$

This is a non-interacting Hamiltonian, whose eigenfunctions are Slater determinants. The form of this equation is reminiscent of the situation in Density functional theory, where the Kohn-Sham equations give a self-consistent field equation with an effective potential $V_{eff}(x)$. We will therefore consider searching for approximate eigenfunctions $|\psi_E\rangle$ of the Coulomb operator $\hat{V}_{ee}$ such that the effective potential is given by $V_{eff}(x) = V(x) + E_{ee}(x)$, where these equations are solved self-consistently. Clearly the full solution will not be a Slater determinant, but in the spirit of Density functional theory we will try to reproduce the electron density generated by the Slater determinant to approximate the effective potential $V_{eff}(x)$.

4.1 Approximation of the effective potential

In order to analyze equation (80), we will use the identities developed in the previous sections. We will focus on the identities involving the Dirac operator, because there are some mathematical properties of these equations that lead to useful simplifications. To this end, we can use equation (73) to rewrite the eigenvalue equation as

$$\int_{-\infty}^{\infty} ds s \hat{K}_{\frac{1}{2}}(s) \, |\psi_E\rangle = \int d^3x E_{ee}(x) \hat{\rho}(x) \, |\psi_E\rangle \, . \tag{83}$$

Define the operator $|\hat{\partial}|$ by

$$|\hat{\partial}| = \int d^3x \hat{a}^\dagger_\sigma(x) |\partial| \hat{a}_\sigma(x), \tag{84}$$

which, written in momentum space, is equal to

$$|\hat{\partial}| = \int d^3p \, \hat{a}^\dagger_\sigma(p) |p| \hat{a}_\sigma(p). \tag{85}$$

Before analyzing this equation, we will first analyze the simpler equation

$$-\int_{-\infty}^{\infty} ds \hat{K}_{\frac{1}{2}}(s, x) \, |\phi\rangle = \int d^3x E_{ee}(x) \hat{\rho}(x) \, |\psi_E\rangle \, , \tag{86}$$

which is the analogous eigenvalue equation for the operator $\hat{G}_{\frac{1}{2}}(x)$.

To analyze these equations, we first define the operator

$$\hat{\not{\partial}} = \int d^3x \, a^\dagger_\sigma(x) \not{\partial} a_\sigma(x). \tag{87}$$

A short calculation shows that the operator $\hat{K}_{\frac{1}{2}}(s, x, y)$ satisfies the equation

$$\frac{\partial}{\partial s} \hat{K}_{\frac{1}{2}}(s, x) = \left[\hat{\not{\partial}}, \hat{K}_{\frac{1}{2}}(s, x) \right], \tag{88}$$

which is the operatorial generalization of equation (20). Define a truncated Green's function by the equation

$$\hat{G}_{\frac{1}{2}}(x;s) = -\int_{-\infty}^{s} d\tilde{s}\hat{K}_{\frac{1}{2}}(\tilde{s},x) - \int_{s}^{\infty} d\tilde{s}\hat{K}_{\frac{1}{2}}(\tilde{s},x). \tag{89}$$

truncated Green's function

$$\int_{-\infty}^{\infty} ds\hat{K}_{\frac{1}{2}}(s,x)\left|\phi\right\rangle = -g_{\frac{1}{2}}(x)\left|\phi\right\rangle, \tag{90}$$

Furthermore, if we define the operator

$$\hat{\not\partial} = \int d^3x\, a^{\dagger}_{\sigma}(x)\not\partial a_{\sigma}(x), \tag{91}$$

then we can write an operatorial generalization of equation (20)

$$\frac{\partial}{\partial s}\hat{K}_{\frac{1}{2}}(s,x) = \left[\hat{\not\partial}, \hat{K}_{\frac{1}{2}}(s,x)\right]. \tag{92}$$

The operator $\hat{K}(s,x)$ has no corresponding equation, which was the reason for introducing the Dirac operator.

These identities are interesting in their own right, and it is likely that there are multiple ways to apply them in a variety of contexts. Here we will use these identities to estimate the correlation energy that appears in density functional theory, with a fairly strong physical approximation.

A Proof of identity

$$|\hat{\partial}| = \int d^3p\, \hat{a}^{\dagger}_{\sigma}(p)|p|\hat{a}_{\sigma}(p). \tag{93}$$

A different way of writing this operator is

$$|\hat{\partial}| = \lim_{s\to 0}\int d^3x d^3y K_{\frac{1}{2}}(s,x,y)a^{\dagger}_{\sigma}(x)a_{\sigma}(y). \tag{94}$$

$$\hat{K}_{\frac{1}{2}}(s) = \int d^3x \int d^3y K_{\frac{1}{2}}(s,x,y) : \hat{\rho}(x)\hat{\rho}(y) :, \tag{95}$$

In Fourier space $\hat{K}_{\frac{1}{2}}(s)$ is given by

$$\hat{K}_{\frac{1}{2}}(s) = \int d^3q \int d^3p d^3p' \tilde{K}_{\frac{1}{2}}(s,p,p') : \hat{\rho}(x)\hat{\rho}(y) :, \tag{96}$$

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